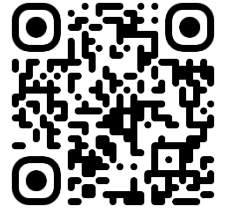


Charles T. Clark

Engineering Portfolio: www.charlesclark.me

charlesclark@vt.edu | (224) 358-5132 | Chicago, IL (Open to Relocation)

Engineering Portfolio



OBJECTIVE

Third-year mechanical engineering student seeking co-op for Spring 2027 and internship for Summer 2027

EDUCATION

Virginia Tech

Bachelor of Science in Mechanical Engineering

Blacksburg, VA

Expected Graduation: May 2028

- GPA: 3.84 | Dean's List (All Semesters)
- Mechanical Engineering Undergraduate Student Ambassador
- Raymond & Shirley Lynn Merit Scholarship Recipient
- Relevant Courses: Dynamics, Thermodynamics, Differential Eq, Mechanics of Bodies, Electrical Theory

EXPERIENCE & PROJECTS

Culligan International

Mechanical Engineering Design Intern

May 2026 – August 2026

Rosemont, IL

- Used SolidWorks and FEA to design, develop and build water treatment components
- 3D modeled and drafted GD&T drawings for an injection-molded float to reduce costs by \$15,000
- Developed booster pump kit to increase water pressure by 150% for commercial hydraulic valve
- Fabricated and sourced a portable field training prototype that secured \$75,000 in funding
- Modeled 3D systems and drafted technical engineering drawings for 5 commercial water softeners

Pulse Jet Design Team

Founder and President

May 2025 – Present

Blacksburg, VA

- Founded a design team with 8 members that prototypes and builds pulse jet engines
- Designed in SolidWorks and fabricated a steel, propane-fueled pulse jet that produced approximately 200 Newtons of peak static thrust during bench testing
- Created a testing bench to safely and accurately get 3 metrics: thrust, temperature, and fuel efficiency

Virginia Tech Engineering Education Department

Accessibility Design Project Leader

August 2024 – May 2025

Blacksburg, VA

- Led a team to design and prototype a telescoping crutch, reducing storage space by 66%
- Utilized CAD and rapid prototyping to create models and show the device's capabilities

Automotive Restoration

Independent Project

August 2020 – Present

Libertyville, IL

- Rebuilt and restored two vehicles, a 2001 Jeep Cherokee and a 1987 Jeep Comanche
- Applied welding, machining, and diagnostic problem solving to restore vehicles to working conditions

CERTIFICATIONS & HONORS

Certified SolidWorks Professional (CSWP) | Dassault Systèmes

July 2026

Finite Element Analysis (FEA) Certification | GoEngineer

July 2026

Eagle Scout | Boy Scouts of America

July 2023

TECHNICAL SKILLS

CAD & Analysis: SolidWorks (CSWP), Finite Element Analysis, Engineering Drawings (GD&T), MATLAB

Product Development: Rapid Prototyping, Injection Molded Part Design, Design Validation

Manufacturing & Fabrication: MIG/Stick Welding, 3D Printing, CNC Machining, Forklift Certification